
EDUCATION

University of Southern California – Los Angeles, CA

- **M.S. Computer Engineering (2024) - GPA: 4.0**
- **B.S. Electrical and Computer Engineering (2023) - GPA: 3.93**
 - Honors: Presidential Scholarship, Summa Cum Laude
 - Relevant Coursework: Computational Intelligence and Neural Learning, Internet and Cloud Computing, Computer Systems Organization, SOC Design, Deep Learning, Probability Theory, Linear Algebra

RESEARCH

Directed Research – Los Angeles, CA

Advised by Dr. Viktor K. Prasanna | Feb 2023 – May 2024

- First Author on **ME-ViT: A Single-Load Memory-Efficient FPGA Accelerator for Vision Transformers** (HiPC 2023 – **Best Paper Award**)
- Designed a Vision Transformer accelerator for FPGAs, achieving up to a 17.89x reduction in memory bandwidth and a 2.16x improvement in throughput per DSP over SOTA designs

Computational Intelligence and Neural Learning Final Project – EE 689 | Aug 2023 – Dec 2023

- Developed a novel method for generating adversarial attacks to induce specific hallucinations for Llama 2 7B in Python
- Implemented reverse backpropagation to calculate embedding vectors for targeted outputs and performed nearest-token searches to identify adversarial input sequences

EXPERIENCE

Apple – Cupertino, CA

FPGA Architecture Intern | May 2024 – Dec 2024

- Developed Python driver and Verilog AXI interface for SPI DAC peripheral on MIX FPGA
- Integrated DisplayPort IP at 8k 60 Hz over CIO80 on Agilex7 FPGA with hardware testing platform
- Designed SERDES link training module and PRBS generator in Verilog for 1.5 GHz loopback tests

AMD/Xilinx – San Jose, CA

RTL Integration Intern | May 2023 – Aug 2023

- Built a distributed design rule verification system (1000+ jobs) in Python to analyze full-chip RTL.

SOC Integration Intern | May 2022 – Aug 2022

- Designed and integrated Python and Perl tools for automated EMIR reporting and simulation processes.

Really, Inc. – Mountain View, CA

Backend Development Intern | Jun 2019 – Mar 2020

- Engineered data preprocessing pipelines for a machine learning platform analyzing social media and email data to detect early signs of Alzheimer's disease
- Built a linguistic analysis tool using BERT, spaCy, and NLTK to measure sentiment shifts and lexical complexity over time, storing results in PostgreSQL for long-term tracking

USC Rocket Propulsion Laboratory – Los Angeles, CA

Lead FPGA Engineer | Sep 2019 – Jan 2023

- Architect for a 13 channel data collection and processing system using an Intel MAX10 FPGA
- Designed a custom float processing ALU for real-time quaternion integration of high throughput data
- Developed Python compiler to generate ROM-based instructions for the processing unit
- Wrote Verilog modules for SD Bus, I2C, and SPI interfaces for ADCs, IMUs, and off-chip memories

PCB Designer

- Designed a high-speed PCB for the FPGA platform, incorporating both analog and digital circuits
- Led development a new battery management PCB with monitoring, charging, and safety features
- Wrote C++ drivers to interface with the FPGA and battery management boards

PERSONAL PROJECTS

Live GPT | Jan 2024 – Current

- Fine-tuned GPT-4o implementation processing live transcription and event data for real-time, conversational interactions
- Wrote retraining algorithm that scores responses and generates self-correcting fine-tuning examples
- Implemented a memory compression technique summarizing key conversation events for efficient context utilization

Quadcopter | Dec 2016 – Jun 2019

- Designed and assembled a self-stabilizing quadcopter with integrated camera streaming
- Wrote all embedded software, including flight controller with PID for stability, and sensor drivers for serial communication

3D Physics Engine | Nov 2019

- Built GPU-accelerated real-time physics and rendering engine in Java using LWJGL
- Designed custom collision detection algorithm suitable for arbitrary object models

LEADERSHIP

USC Climbing Team Captain | Aug 2021 – May 2024

- Led weekly team practices and develop group workouts for competition training
- Mentored members on technique, strength training, and injury rehab

Los Altos Hacks Organizer | Jan 2017 – Jun 2019

- Contacted companies for scholarships and helped the team raise over \$25,000 for the event
- Worked with a team of 12 students to secure funding, a venue, judges, and participants

OTHER ACTIVITIES

Eagle Scout	2018
SM Hacks Best Web Application	2017
Code Day Hackathon Best Web Application	2016
Rock Climbing, Guitar, Kendama	